

Ascorbic-acid transporter Slc23a1 is essential for vitamin C transport into the brain and for perinatal survival

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The only proven requirement for ascorbic acid (vitamin C) is in preventing scurvy, presumably because it is a cofactor for hydroxylases required for post-translational modifications that stabilize collagen. We have created mice deficient in the mouse ortholog (solute carrier family 23 member 1 or Slc23a1) of a rat ascorbic-acid transporter, Svct2 (ref. 4). Cultured embryonic fibroblasts from homozygous Slc23a1 / mice had less than 5% of normal ascorbic-acid uptake. Ascorbic-acid levels were undetectable or markedly reduced in the blood and tissues of Slc23a1 / mice. Prenatal supplementation of pregnant females did not elevate blood ascorbic acid in Slc23a1 / fetuses, suggesting Slc23a1 is important in placental ascorbic-acid transport. Slc23a1 / mice died within a few minutes of birth with respiratory failure and intraparenchymal brain hemorrhage. Lungs showed no postnatal expansion but had normal surfactant protein B levels. Brain hemorrhage was unlikely to be simply a form of scurvy since Slc23a1 / mice showed no hemorrhage in any other tissues and their skin had normal skin 4-hydroxyproline levels despite low ascorbic-acid content. We conclude that Slc23a1 is required for transport of ascorbic acid into many tissues and across the placenta. Deficiency of the transporter is lethal in newborn mice, thereby revealing a previously unrecognized requirement for ascorbic acid in the perinatal period.